SCM820

Command Strings

Third-party commands for Shure SCM820

Version: 3.1 (2021-D)

Table of Contents

SCM820 Command Strings **3**

Command Strings for External Controller **3**

Channels 3

Device Commands **3**

Audio Commands 5

Channel Commands 7

Digital Signal Processing Commands 8

Monitoring Commands 12

Error Codes 15

SCM820
Command Strings

Command Strings for External Controller

The SCM820 mixer can be connected via Ethernet to a control system, such as AMX®, Crestron® or Extron®.

- Connection: Ethernet (TCP/IP; select “Client” in the AMX/Crestron program)
- Port: 2202

The SCM820 has 4 types of strings, as follows:

1. GET – The GET command is used to find the status of a parameter. After the AMX/Crestron sends a GET command, the SCM820 responds with a REPORT string.
2. SET – The SET command is used to change the status of a parameter. After the AMX/Crestron sends a SET command, the SCM820 will respond with a REPORT string to indicate the new value of the parameter.
3. REP – When the SCM820 receives a GET or SET command, it will reply with a REPORT command to indicate the status of the parameter. REPORT is also sent by the MXW System when a parameter is changed via the front panel or via the GUI.
4. SAMPLE – Used for metering audio levels.

All messages sent and received are ASCII. Level indicators and gain indicators are also in ASCII. The SCM820 will send a REPORT command when any of these parameters change.

Note: This document is the most popular commands. There are many more commands available. If you need to control/monitor other parameters, please contact us at support@shure.com and we will be able to assist you.

A Crestron macro (written by Shure) can be downloaded here.

Channels

The character “x” in all of the following strings represents the channel of the mixer and can be ASCII numbers 0 through 19 as in the following table:

ASCII Character	SCM820 Channel
0	All Channels
1 through 8	Input Channels
9	Aux Input
10 through 17	Direct Outputs
18	Output A
19	Output B

Device Commands

View Device ID	Command String:	< GET DEVICE_ID >	*The Device ID command does not contain the x channel character, as it is for the entire device.*
	SCM820 Response:	< REP DEVICE_ID {yyyyyyyyyyyyyyyyyyyyyyyyyyyyyyy} >	*Where yyyyyyyyyyyyyyyyyyyyyyyyyyyyyyy is 31 characters of the device ID. The SCM820 always responds with a 31 character device ID.*
Set Device ID	Command String:	< SET DEVICE_ID {yyyyyyyy} >	*Where yyyyyyyy is 31 characters of the device ID. The device ID can be 1 to 31 characters long.*
	SCM820 Response:	< REP DEVICE_ID {yyyyyyyyyyyyyyyyyyyyyyyyyyyyyyy} >	*Where yyyyyyyyyyyyyyyyyyyyyyyyyyyyyyy is 31 characters of the device ID. The SCM820 always responds with a 31 character device ID.*
Flash Lights on SCM820	Command String:	< SET FLASH ON > < SET FLASH OFF >	*Send one of these commands to the SCM820. The flash automatically turns off after 60 seconds.*
	SCM820 Response:	< REP FLASH ON > < REP FLASH OFF >	*The SCM820 will respond with one of these strings.*
Set Auto Link Mode	Command String:	< SET AUTO_LINK_MODE ON > < SET AUTO_LINK_MODE OFF >	*This set of commands forces a channel to be ON. If this is OFF, the channel would typically be in Automix mode.* *Mix A and Mix B have different commands. Send one of these commands, depending on the mix.*
	SCM820 Response:	< REP AUTO_LINK_MODE ON > < REP AUTO_LINK_MODE OFF >	*The SCM820 will respond with one of these strings.*
Get Auto Link Mode	Command String:	< GET AUTO_LINK_MODE >	
	SCM820 Response:	< REP AUTO_LINK_MODE ON >	*The SCM820 will respond with one of these strings.*

		< REP AUTO_LINK_MODE OFF >	

Audio Commands

Get Audio Gain	Command String:	< GET x AUDIO_GAIN_HI_RES >	*Where x is ASCII channel number: 0, 1-9, and 18-19 (see Channels table).*
	SCM820 Response:	< REP x AUDIO_GAIN_HI_RES yyyy >	*Where yyyy takes on the ASCII values of 0000 to 1280. yyyy is in steps of one-tenth of a dB.*
Set Audio Gain	Command String:	< SET x AUDIO_GAIN_HI_RES yyyy >	*Where yyyy takes on the ASCII values of 0000 to 1280. yyyy is in steps of one-tenth of a dB.*
	SCM820 Response:	< REP x AUDIO_GAIN_HI_RES yyyy >	*Where yyyy takes on the ASCII values of 0000 to 1280.*
Increase Audio Gain by *n* dB	Command String:	< SET x AUDIO_GAIN_HI_RES INC nn >	*Where nn is the amount in one-tenth of a dB to increase the gain. nn can be single digit (n), double digit (nn), triple digit (nnn).*
	SCM820 Response:	< REP x AUDIO_GAIN_HI_RES yyyy >	*Where yyyy takes on the ASCII values of 0000 to 1280.*
Decrease Audio Gain by *n* dB	Command String:	< SET x AUDIO_GAIN_HI_RES DEC nn >	*Where nn is the amount in one-tenth of a dB to decrease the gain. nn can be single digit (n), double digit (nn), triple digit (nnn).*
	SCM820 Response:	< REP x AUDIO_GAIN_HI_RES yyyy >	*Where yyyy takes on the ASCII values of 0000 to 1280.*
Get Audio Mute	Command String:	< GET x AUDIO_MUTE >	*Where x is ASCII channel number: 0, 1-9, and 18-19 (see Channels table).*

	SCM820 Response:	< REP x AUDIO_MUTE ON > < REP x AUDIO_MUTE OFF >	*The SCM820 will respond with one of these strings.*
Mute Audio	Command String:	< SET x AUDIO_MUTE ON >	
	SCM820 Response:	< REP x AUDIO_MUTE ON >	
Unmute Audio	Command String:	< SET x AUDIO_MUTE OFF >	
	SCM820 Response:	< REP x AUDIO_MUTE OFF >	
Toggle Audio Mute	Command String:	< SET x AUDIO_MUTE TOGGLE >	
	SCM820 Response:	< REP x AUDIO_MUTE ON > < REP x AUDIO_MUTE OFF >	*The SCM820 will respond with one of these strings.*
Input Audio Mix Bus	Command String:	< SET x INPUT_AUDIO_MIX_BUS sts > < GET x INPUT_AUDIO_MIX_BUS >	*Where x is the channel and takes on value 0 and 1 - 8 for input channels and 9 for aux input* *Where sts represents the desired setting and takes on values:* *MIXBUS_A* *MIXBUS_B* *BOTH* *NEITHER*

	SCM820 Response:	< REP **x** INPUT_AUDIO_MIX_BUS sts > < REP ERR >	*The SCM820 will respond with one of these strings.*

Channel Commands

View Channel Name	Command String:	< GET x CHAN_NAME >	*Where x is ASCII channel number: 0 through 19.*
	SCM820 Response:	< REP x CHAN_NAME {yyyyyyyyyyyyyyyyyyyyyyyyyyyyyyy} >	*Where yyyyyyyyyyyyyyyyyyyyyyyyyyyyyyy is 31 characters of the user name. The SCM820 always responds with a 31 character name.*
Set Channel Name	Command String:	< SET x CHAN_NAME {yyyyyyyy} >	*Where yyyyyyyy is 31 characters of the channel name. The channel name can be 1 to 31 characters long. Each channel must have a unique name.*
	SCM820 Response:	< REP x CHAN_NAME {yyyyyyyyyyyyyyyyyyyyyyyyyyyyyyy} >	*Where yyyyyyyyyyyyyyyyyyyyyyyyyyyyyyy is 31 characters of the channel name. The SCM820 always responds with a 31 character name.*
Get Channel Forced On	Command String:	< GET x ALWAYS_ON_ENABLE_A > < GET x ALWAYS_ON_ENABLE_B >	*This set of commands forces a channel to be ON. If this is OFF, the channel would typically be in Automix mode.* *Mix A and Mix B have different commands. Send one of these commands, depending on the mix.*
	SCM820 Response:	< REP x ALWAYS_ON_ENABLE_A OFF > < REP x ALWAYS_ON_ENABLE_B OFF > < REP x ALWAYS_ON_ENABLE_A ON > < REP x ALWAYS_ON_ENABLE_B ON >	*The SCM820 will respond with one of these strings.*

Set Channel Forced On	Command String:	< SET x ALWAYS_ON_ENABLE_A OFF > < SET x ALWAYS_ON_ENABLE_A ON > < SET x ALWAYS_ON_ENABLE_A TOGGLE > < SET x ALWAYS_ON_ENABLE_B OFF > < SET x ALWAYS_ON_ENABLE_B ON > < SET x ALWAYS_ON_ENABLE_B TOGGLE >	*This set of commands forces a channel to be ON. If this is OFF, the channel would typically be in Automix mode.* *Mix A and Mix B have different commands. Send one of these commands, depending on the mix.*
	SCM820 Response:	< REP x ALWAYS_ON_ENABLE_A OFF > < REP x ALWAYS_ON_ENABLE_B OFF > < REP x ALWAYS_ON_ENABLE_A ON > < REP x ALWAYS_ON_ENABLE_B ON >	*The SCM820 will respond with one of these strings.*
Input Audio Source	Command String:	< GET x INPUT_AUDIO_SOURCE > < SET y INPUT_AUDIO_SOURCE **source** >	*Where x is channel number 0 through 9* *Where y is channel number 1 through 9* *Where **source** indicates the desired source:* *Analog* *Network*
	SCM820 Response:	< REP y INPUT_AUDIO_SOURCE **source** >	*Returns an error for channels 10 to 19*

Digital Signal Processing Commands

Set IntelliMix® Mode	Command String:	< SET x INTELLIMIX_MODE CLASSIC >	*Where x is ASCII channel number: 18-19.*

		< SET x INTELLIMIX_MODE SMOOTH > < SET x INTELLIMIX_MODE EXTREME > < SET x INTELLIMIX_MODE CUSTOM > < SET x INTELLIMIX_MODE MANUAL > < SET x INTELLIMIX_MODE CUSTOM_PRESET >	*Send only one of these commands.*
	SCM820 Response:	< REP x INTELLIMIX_MODE CLASSIC > < REP x INTELLIMIX_MODE SMOOTH > < REP x INTELLIMIX_MODE EXTREME > < REP x INTELLIMIX_MODE CUSTOM > < REP x INTELLIMIX_MODE MANUAL > < REP x INTELLIMIX_MODE CUSTOM_PRESET >	*The SCM820 will respond with one of these strings.*
Get IntelliMix Mode	Command String:	< GET x INTELLIMIX_MODE >	*Where x is ASCII channel number: 18-19*
	SCM820 Response:	< REP x INTELLIMIX_MODE CLASSIC > < REP x INTELLIMIX_MODE SMOOTH > < REP x INTELLIMIX_MODE EXTREME > < REP x INTELLIMIX_MODE CUSTOM > < REP x INTELLIMIX_MODE MANUAL > < REP x INTELLIMIX_MODE CUSTOM_PRESET >	*The SCM820 will respond with one of these strings.*

Set Assign DFR 1	Command String:	< SET DFR1_ASSIGNED_CHAN x >	*Where x is ASCII channel number: 1-8, 18-19 (see Channels table). x is 20 for Unassigned.*
	SCM820 Response:	< REP DFR1_ASSIGNED_CHAN xxx >	*Where x is ASCII channel number: 001-008, 018-019 (see Channels table). x is 020 for Unassigned.*
Set Assign DFR 2	Command String:	< SET DFR2_ASSIGNED_CHAN x >	*Where x is ASCII channel number: 1-8, 18-19 (see Channels table). x is 20 for Unassigned.*
	SCM820 Response:	< REP DFR2_ASSIGNED_CHAN xxx >	*Where x is ASCII channel number: 001-008, 018-019 (see Channels table). x is 020 for Unassigned.*
Get Assign DFR 1	Command String:	< GET DFR1_ASSIGNED_CHAN >	
	SCM820 Response:	< REP DFR1_ASSIGNED_CHAN xxx >	*Where x is ASCII channel number: 001-008, 018-019 (see Channels table). x is 020 for Unassigned.*
Get Assign DFR 2	Command String:	< GET DFR2_ASSIGNED_CHAN >	
	SCM820 Response:	< REP DFR2_ASSIGNED_CHAN xxx >	*Where x is ASCII channel number: 001-008, 018-019 (see Channels table). x is 020 for Unassigned.*
Set DFR 1 Bypass	Command String:	< SET DFR1_BYPASS ON > < SET DFR1_BYPASS OFF >	*Send one of these commands to the SCM820.*
	SCM820 Response:	< REP DFR1_BYPASS ON > < REP DFR1_BYPASS OFF >	*The SCM820 will respond with one of these strings.*
Set DFR 2 Bypass	Command String:	< SET DFR2_BYPASS ON > < SET DFR2_BYPASS OFF >	*Send one of these commands to the SCM820.*

	SCM820 Response:	< REP DFR2_BYPASS ON > < REP DFR2_BYPASS OFF >	*The SCM820 will respond with one of these strings.*
Get DFR 1 Bypass	Command String:	< GET DFR1_BYPASS >	
	SCM820 Response:	< REP DFR1_BYPASS ON > < REP DFR1_BYPASS OFF >	*The SCM820 will respond with one of these strings.*
Get DFR 2 Bypass	Command String:	< GET DFR2_BYPASS >	
	SCM820 Response:	< REP DFR2_BYPASS ON > < REP DFR2_BYPASS OFF >	*The SCM820 will respond with one of these strings.*
Clear DFR 1 Filters	Command String:	< SET DFR1_CLEAR_ALL_FILTERS ON >	*There is no need to send an OFF command*
	SCM820 Response:	< REP DFR1_CLEAR_ALL_FILTERS ON >	
Clear DFR 2 Filters	Command String:	< SET DFR2_CLEAR_ALL_FILTERS ON >	*There is no need to send an OFF command*
	SCM820 Response:	< REP DFR2_CLEAR_ALL_FILTERS ON >	
Set DFR 1 Freeze	Command String:	< SET DFR1_FREEZE ON > < SET DFR1_FREEZE OFF >	*Send one of these commands to the SCM820.*
	SCM820 Response:	< REP DFR1_FREEZE ON > < REP DFR1 FREEZE OFF >	*The SCM820 will respond with one of these strings.*

Set DFR 2 Freeze	Command String:	< SET DFR2_FREEZE ON > < SET DFR2_FREEZE OFF >	*Send one of these commands to the SCM820.*
	SCM820 Response:	< REP DFR2_FREEZE ON > < REP DFR2 FREEZE OFF >	*The SCM820 will respond with one of these strings.*
Get DFR 1 Freeze	Command String:	< GET DFR1_FREEZE >	
	SCM820 Response:	< REP DFR1_FREEZE ON > < REP DFR1_FREEZE OFF >	*The SCM820 will respond with one of these strings.*
Get DFR 2 Freeze	Command String:	< GET DFR2_FREEZE >	
	SCM820 Response:	< REP DFR2_FREEZE ON > < REP DFR2_FREEZE OFF >	*The SCM820 will respond with one of these strings.*

Monitoring Commands

Get Gate Status, Mix A	Command String:	< GET x INPUT_AUDIO_GATE_A >	*Where x is ASCII channel number: 0 through 8 (see Channels table).* *It is not necessary to continually send this command. The SCM820 will send a REPORT message whenever the status changes.*
	SCM820 Response:	< REP x INPUT_AUDIO_GATE_A ON > < REP x INPUT_AUDIO_GATE_A OFF >	*The SCM820 will respond with one of these strings.*
Get Gate Status, Mix B	Command String:	< GET x INPUT_AUDIO_GATE_B >	*Where x is ASCII channel number: 0 through 8 (see Channels table).*

			It is not necessary to continually send this command. The SCM820 will send a REPORT message whenever the status changes.
	SCM820 Response:	< REP x INPUT_AUDIO_GATE_B ON > < REP x INPUT_AUDIO_GATE_B OFF >	*The SCM820 will respond with one of these strings.*
Get Limiter Engaged Status	Command String:	< GET x LIMITER_ENGAGED >	*Where x is ASCII channel number: 0 and 18-19 (see Channels table).* *It is not necessary to continually send this command. The SCM820 will send a REPORT message whenever the status changes.*
	SCM820 Response:	< REP x LIMITER_ENGAGED ON > < REP x LIMITER_ENGAGED OFF >	*The SCM820 will respond with one of these strings.*
Get Input Clip Status	Command String:	< GET x AUDIO_IN_CLIP_INDICATOR >	*Where x is ASCII channel number: 0 through 9 (see Channels table).* *It is not necessary to continually send this command. The SCM820 will send a REPORT message whenever the status changes.*
	SCM820 Response:	< REP x AUDIO_IN_CLIP_INDICATOR ON > < REP x AUDIO_IN_CLIP_INDICATOR OFF >	*The SCM820 will respond with one of these strings.*

Get Output Clip Status	Command String:	< GET x AUDIO_OUT_CLIP_INDICATOR >	*Where x is ASCII channel number: 10 through 19 (see Channels table).* *It is not necessary to continually send this command. The SCM820 will send a REPORT message whenever the status changes.*
	SCM820 Response:	< REP x AUDIO_OUT_CLIP_INDICATOR ON > < REP x AUDIO_OUT_CLIP_INDICATOR OFF >	*The SCM820 will respond with one of these strings.*
Turn Metering On	Command String:	< SET METER_RATE sssss >	*Where sssss is the metering speed in milliseconds.* *Setting sssss=0 turns metering off. Minimum setting is 100 milliseconds. Metering is off by default.*
	SCM820 Response:	< REP METER_RATE sssss > < SAMPLE aaa bbb ccc ddd eee fff ggg hhh iii jjj kkk lll mmm nnn ooo ppp qqq rrr sss >	*Where aaa, bbb, etc is the value of the audio level received and is 000-120.* *aaa = input 1* *bbb = input 2* *ccc = input 3* *ddd = input 4* *eee = input 5* *fff = input 6* *ggg = input 7* *hhh = input 8* *iii = aux input* *jjj = direct out 1* *kkk = direct out 2*

			lll = direct out 3 *mmm = direct out 4* *nnn = direct out 5* *ooo = direct out 6* *ppp = direct out 7* *qqq = direct out 8* *rrr = mix out A* *sss = mix out B*
Stop Metering	Command String:	< SET METER_RATE 0 >	*A value of 00000 is also acceptable.*
	SCM820 Response:	< REP METER_RATE 00000 >	

Error Codes

There is a < REP ERR > error string that indicates the command is not able to be implemented. This is usually due to a typo or a command that does not exist.